



CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)
BSc CHEMISTRY

Programme	B. Sc Chemistry				
Course Title	INORGANIC CHEMISTRY-V				
Type of Course	MAJOR				
Semester	VII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Theories of acids and bases, Bonding in diborane, Atomic number, mass number, isotopes, Basic idea about nuclear radioactivity, Nuclear fission and fusion, Basic concepts in Coordination Chemistry, Knowledge about magnetic properties of complexes				
Course Summary	This course enables the students to develop knowledge about HSAB principle, Concepts of superacids, Bonding and structure in higher boranes. It deals with detailed nuclear shell models, Nuclear reactions, Neutron activation analysis and Radiation chemistry It covers the electronic spectra of complexes and explanation of d-d transition It gives understanding about the various magnetic properties and its calculation It enables the students to familiarise with various spectral techniques that is used to characterise the metal complexes like ESR, NMR and Mossbauer It covers the different types of reactions that takes place in complexes and their explanations This course enables the students to apply knowledge in the qualitative analysis of mixture of ions and develop skill for separation and estimation of mixture of ions				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate a deep understanding of HSAB principle, Super acid concepts and bonding structure in higher boranes and boron cluster compounds.	U	C	Instructor-created exams / Assignments/Quiz
CO2	Explain detailed nuclear shell models, nuclear reactions, neutron activation analysis and radiation chemistry.	U	C	Instructor-created exams / Assignments/ seminar presentations
CO3	Comprehend electronic spectra of complexes and understand various magnetic properties of complexes.	U	C	Class test /Assignment / seminar/Quiz
CO4	Grasp the reactions of coordination compounds, the mechanisms and apply theoretical understandings.	Ap	C	Assignment/Seminar/Test
CO5	To understand chemistry of excited state coordination compounds	U	C	Assignment/Seminar/ Test
CO6	Apply practical skills in the analysis of mixture of metal ions, specifically rare earth elements and separation and estimation of binary mixture of ions in solution.	An	P	Viva Voce/Observation of practical skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+30)	Marks
I	ACID BASE CONCEPTS AND COMPOUNDS OF BORON		10	21
	1	Classification of acids and bases as Hard and Soft	1	
	2	HSAB principle. The theoretical basis of hardness and softness	1	
	3	The Drago-Wayland equation, E and C parameters- Symbiosis, Applications of HSAB concept, Super acids	2	
	4	Electron-deficient compounds- Boron hydrides- Preparation, Reactions, Structure, and Bonding. Styx numbers-Closo, nido, arachno polyhedral structures.	3	
	5	Boron cluster compounds-Wade's rule, Polyhedral borane anion- Carboranes, Metallaboranes and Metallacarboranes.	3	
II	NUCLEAR AND RADIATION CHEMISTRY		7	16
	6	Structure of nucleus: Shell, liquid drop, Fermi gas, Collective and Optical models. Nuclear reaction: Bethe's notation of nuclear process	2	
	7	Types- Reaction cross section, Photonuclear and Thermonuclear reactions. Nuclear fission: Theory of fission, Neutron capture cross section and Critical size, Nuclear fusion.	2	
	8	Neutron activation analysis.	1	
	9	Detection and measurement of radiation- GM and Scintillation counters, Radiolysis of water, Radiation hazards, Radiation dosimetry.	2	
III	SPECTRAL AND MAGNETIC PROPERTIES OF COORDINATION COMPOUNDS		18	40
	10	Spectroscopic ground state, Terms for d^n configuration, Selection rule for d-d transition, Effect of ligand field on R S terms, O_h and T_h complexes.	2	

	11	Orgal diagram, Spectra of 3d (d^1, d^2, d^3) metal ions complexes, Racah parameter, Charge transfer parameter (LMCT. MLCT) with example.	2	
	12	Types of Magnetic properties: Paramagnetism and Diamagnetism, Curie and Curie- Weiss laws. The μ_J , μ_{L+S} , and μ_S expressions.	2	
	13	Orbital contribution to magnetic moment and its quenching, Spin-orbit coupling, Temperature independent Paramagnetism, Anti ferromagnetism- Types and exchange pathways. Determination of magnetic moment by Gouy method.	3	
	14	ESR spectra – Application to copper complexes.	3	
	15	NMR spectroscopy for structural studies of diamagnetic metal complexes from chemical shift and spin- spin coupling.	3	
	16	Mossbauer spectroscopy- the Mossbauer Effect, Hyperfine interactions (qualitative treatment). Application to Iron and Tin compounds	3	
	IV REACTIONS OF COORDINATION COMPOUNDS		10	21
	17	Ligand substitution reactions, Labile and Inert complexes, Rate law, Classification of mechanisms- D, A and I mechanisms. Substitution reactions in Octahedral complexes.	2	
	18	The Eigen-Wilkins Mechanism. Fuoss-Eigenequation. Aquation and base hydrolysis- Mechanism. Substitution reactions in square planar complexes	2	
	19	The Trans effect- Applications and theories of Trans effect, The cis effect.	1	
	20	Classification of redox reaction mechanisms. Outer sphere and Inner sphere mechanisms, Marcus equation, Effect of the bridging ligand.	1	
	21	Methods for distinguishing outer- and inner-sphere redox reactions, Photochemical reactions of metal complexes.	2	
	22	Prompt and delayed reactions, Excited states of metal complexes- Inter ligand, ligand field, charge transfer, and delocalized states Properties of ligand field excited states. Photosubstitution- Prediction of substitution lability by Adamson`s rules.	2	

V	QUALITATIVE MIXTURE ANALYSIS INCLUDING LESS COMMON CATIONS and ESTIMATION OF BINARY MIXTURES		15*2 (30)	
	23	1. Inorganic Cation Mixture Analysis Separation and identification of four metal ions including less common elements like W, Se Te, Mo, Ce, , Zr, V, and Li. (Eliminating acid radicals not present). Confirmation by Spot tests. 0. Estimation of ions in mixture (Open ended) a) Separation and estimation of binary mixtures of ions in solution [Cu^{2+}, Ni^{2+}, Fe^{2+}, Ca^{2+}, Mg^{2+} and $(\text{Cr}_2\text{O}_7)^{2-}$] by volumetric, colorimetric or gravimetric methods. Only one of the components to be estimated. Any two combinations can be performed		

REFERENCES:

1. N.N. Greenwood and A. Earnshaw, *Chemistry of Elements*, 2/e, Elsevier Butterworth-Heinemann, 2005.
2. J. E. Huheey, E. A. Keiter, R. L. Keiter. O.K. Medhi. *Inorganic Chemistry, principles of structure and reactivity*, Pearson Education, 2006.
3. G.L. Miessler, D.A. Tarr, *Inorganic Chemistry*, Pearson, 2010.
4. D.F. Shriver, P.W. Atkins, *Inorganic Chemistry*, Oxford University Press, 2002
5. William W Porterfield, *Inorganic Chemistry-Aunified approach*, Academic Press, 2005.
6. Keith F Purcell, John C Kotz, *Inorganic Chemistry*, Cengage Learning, 2010.
7. James E House, *Inorganic Chemistry*, Academic Press, 2008.
8. H.J. Arnikar, *Essentials of Nuclear chemistry*, New Age International, 2005.
9. Friedlander and J.W. Kennedy, *Introduction to Radiochemistry*, John Wiley and Sons, 1981.

- 10.S. Glasstone, *Source Book on Atomic Energy*, 3rd edn., Affiliated East-West Press Pvt.Ltd., 1967.
11. B.Douglas, D.McDaniel, J.Alexander, *Concepts and Models of Inorganic Chemistry*, Wiley Student Edition, 2006.
12. A.W.Adamson and P.D.Fleischauer, *Concepts of Inorganic Photochemistry*, Wiley.
13. F.A.Cotton and G.Wilkinson, *Advanced Inorganic Chemistry*, Wiley.
14. A.Earnshaw, *Introduction to Magnetochemistry*, Academic Press, 1968.
- 15 R.L.Dutta and A.Shyamal, *Elements of Magnetochemistry*, S.Chand and Co.1982.
16. A.E. Martell, *Coordination Chemistry*, Vol.I
17. R.S. Drago, *Physical Methods in Inorganic Chemistry*, Affiliated East- West Press Pvt. Ltd.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O 5	PS O 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	-	-	3	2	3	2	2	-	3	-	1
CO 2	3	2	-	-	3	2	3	2	1	-	3	-	1
CO 3	3	2	-	-	2	2	3	2	1	-	3	-	2
CO 4	3	2	-	-	2	3	3	2	1	-	3	-	1
CO 5	3	2	-	1	2	3	3	2	1	-	3	-	1
CO 6	2	-	3	3	3	3	3	2	1	2	3	2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment /viva	Practical skill evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓